

Ph.D Thesis Introductory Chapter

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Contents

1	Overview	1
2	t-SNE and Dimension Reduction	2
2.1	The Ubiquity and Challenge of High-Dimensional Data	2
2.2	A Brief History of Dimensionality Reduction	4
2.3	The Road to t-SNE	6

1 Overview

As we venture further into the 21st century, our world is shaped, built, or dare we say, defined by data (Hey et al. [2009]). From simple information like class rosters, to large scale datasets like the Human Genome Project, we are recording everything, with the realization that there are limitless values to be extracted. Mathematics and algorithms are at the center of this grand picture, as the extent to which we can unlock this value is fundamentally constrained by our mathematical understanding. Indeed, for decades have we collected visual data such as satellite imagery and medical scans, but the studies and utilization of them remained primarily manual. In 2012, the success of ImageNet(Krizhevsky et al. [2012]), along with Convolution Neural Network (CNN), made automated analysis of visual data possible. It was moments like this that revealed to us there is so much to be seen and discovered with data.

Among all the mathematical theories and tools that forge the foundation of data science, probability theory and probabilistic algorithms are of unique and paramount importance. With data comes uncertainty and noise, and with forecasting and prediction comes ambiguity. The nature of the tasks data science aims to tackle and the intrinsic problems stemming from all forms of data call for probabilistic approaches. It provides the mathematical language and framework for us to navigate through randomness with deterministic confidence.

Take the example of CNN we just mentioned. The success of CNN is seen by many as the watershed moment in the history of mathematics and computer science. Though the advancements in semiconductor industry and hardware engineering should never be overlooked, the birth of CNN is the result of a series of brilliant developments in probability since Kholmogorov. The applications of concepts like softmax layers, maximum log-likelihood estimation, etc all have a deep root in probability theory and constitute the core mechanism

of CNN(Krizhevsky et al. [2012]). And our stories have just begun. As of today, the development of Large Language Models (LLM), the closest ever creation of humans to Artificial General Intelligence (AGI) is essentially a probabilistic algorithm (Vaswani et al. [2017]).

In this thesis, we will explore three different probabilistic algorithms. In chapter 2, we will study t-SNE, another famous and important dimension reduction algorithm developed by Hinton (van der Maaten and Hinton [2008]). Will there still be an equilibrium measure for t-SNE if we swap out its input and output kernels? How does the performance differ? Those are the questions we aim to answer. In chapter 3, we will transform a classic theorem, Lovász Local Lemma, into a probabilistic algorithm that can learn difficult combinatorial structures. We will also provide the theoretical foundation of the algorithm, proving its correctness and giving an estimation of the running time expectation. In chapter 4, we will dive into another dimension reduction algorithm, Variational Autoencoder (VAE), and how a particularly designed VAE can handle missing data via imputation and help us through factor analysis (thus the name VarFA). All three algorithms represent a different yet extremely important aspect of data science, each with its own charm and characteristics.

For the rest of this chapter, we will provide a brief introduction to the probabilistic algorithms we will discuss in this thesis. From basic concepts, to history, and related progress. It is very exciting to see all the mathematical concepts our predecessors have developed, and have been passionate about, are being turned into new technologies that impact and steer the modern world. And may this thesis provides an insignificant, yet meaningful addition to this great painting of our future.

2 t-SNE and Dimension Reduction

2.1 The Ubiquity and Challenge of High-Dimensional Data

Most data and data sets we deal with on a daily basis are multi-dimensional by nature. For example, a simple class roster can have multiple exam and homework scores. If the total number of assignments is greater than 3, then this single class roster is already beyond what we can natively visualize (if we want to represent each student as a point). On the other hand, each image can be seen as a high-dimensional vector and the dimensionalities increase rapidly while the resolution we are able to capture advances with technology. The number of variables scale in a breakneck pace as we are more capable of capturing detail-rich data as well as automating the process (Donoho [2000]). As stated in the last section, this process gives us practically unlimited potentials. The objects and processes we would like to study were recorded with utmost details, unlike the past when vital bits of information could be lost forever. However, this also brings caveats and new mathematical challenges.

One of the most important and hardest challenges is the noise that comes with data. Apart from noise caused by engineering factors, the growth of dimensionality also brings noise because many dimensions may be irrelevant for the problems one want to solve. In 1996, Robert Tibshirani discovered the famous Least Absolute Shrinkage and Selection Operator (LASSO) for the purpose of variable selection (Tibshirani [1996]). Consider a sample consisting of N cases, each with dimension p . Let y_i be the outcome and x_i be the p -dimensional vector that represents the i -th case. LASSO aims to solve

$$\min_{\beta_0, \beta} \sum_{i=1}^N (y_i - \beta_0 - x_i^T \beta)^2, \quad (2.1)$$

with constraint

$$\sum_{j=1}^p |\beta_j| \leq t. \quad (2.2)$$

LASSO marks the early realization and effort mathematicians and scientists were trying to mitigate the overwhelming information brought by high dimensional data. LASSO was also closely related to Basis Pursuit Denoising (BPDN), introduced in 1994 (Chen and Donoho [1994]). BPDN is a well known NP-hard problem. Before the age of big data comes, mathematicians had been thinking about extracting useful signals from noise/spurious correlations. After the dawn of 21st century and the explosive growth of computer science, problems like BPDN and LASSO become even more important after we add computational cost to the equation (Fan and Lv [2010]).

Apart from feature and variable selection, people also seek to “compress” the data, creating a mathematical summary of the essence. t-SNE is one of such examples. Instead of making a selection, t-SNE aims to find a low-dimensional representation of the data and taking the entire data set into account. Another good example is the Variational Auto Encoder (Kingma and Welling [2014]), a generative model that is able to scale well with large data sets using a low-dimensional “latent space”. We will discuss VAE and our related work on this topic in chapter 4.

In the face of large amount of high dimensional data, digesting it “as is” is more often than not impossible. The above examples show us that we either make a selection, or we perform compression, which are the two most common approaches in terms of getting what we need from the data. We would like to discuss this further because high-dimensional data is not only mathematically and computationally hard to deal with, but also tends to mask the insights we are looking for. This is somehow a paradoxical nature of high-dimensionality. Faithful recording of data assures us the information and values we seek are within the data sets and within in our reach. But the form and scaling of the high-dimensional data also buries the light we are after. This is the so-called “curse of dimensionality”.

The term “curse of dimensionality” was first coined by Richard Bellman in 1961 (Bellman [1961]). It refers to various phenomena that appears when analyzing data in high-dimensional space that do not occur when dimensionality is low. Qualitatively speaking, the “curse” usually means that as dimensionality increases, the exponential increase of volume makes the same amount of data sparse and harder to work with.

To understand this concept, let us first look at the famous example of hypercube vs hypersphere. Consider a hypercube with edge length $2r$ in d -dimensional Euclidean space and its inscribed sphere with radius r . Denote Γ the gamma function and C_d , S_d the volume of the hypercube and hypersphere respectively. Then we have

$$\lim_{d \rightarrow \infty} \frac{S_d}{C_d} = \lim_{d \rightarrow \infty} \frac{\pi^{d/2}}{d 2^{d-1} \Gamma(d/2)} = 0. \quad (2.3)$$

This means that our low-dimensional intuition fails. The volume of hypercubes are mostly “concentrated” in the corners and the inscribed sphere’s volume becomes negligible. If we want to analyze the data related to the hypercube, a data set that is uniformly distributed in the hypercube will become an increasingly small part of the whole space. In general, as we increase the dimension of our sample space, any finite data set will occupy a vanishingly small part of it. On the other hand, to maintain the roughly same density, one needs to accumulate exponentially more data points to compensate for the increase in dimensions.

Such fact also induces a data mining problem, where it is unrealistic to gather enough amount of data points to guarantee the separability of data generated by two different distributions. It is a well known fact in machine learning that too many features with inadequate data

points will lead to overfitting. This discovery is not new, either. In 1968, Hughes showed that adding more features past certain points without providing more data points will cause worse classification performance (Hughes [1968]). Later it becomes known as the famous Hughes phenomenon. Adding features is not a free lunch and the results could be detrimental.

With all the above being said, it is natural to consider dimension reduction techniques and algorithms. The knowledge of which aspects of data are important is obviously not a priori, so it is impossible to select the features/dimensions we need beforehand. The only option for us is to develop the mathematical theory to reduce the dimensionality to the extent that the information we seek becomes apparent. Now we are almost a quarter into the 21st century, the development of technology resonates with the need for dimension reduction, and puts our mathematical legacy to the test.

2.2 A Brief History of Dimensionality Reduction

The theory of dimension reduction can be traced back to the beginning of 20th century. In 1901, Karl Pearson introduced the geometric idea of finding lines and planes of best fit to data points, later known as the famous Principle Component Analysis (PCA) (Karl Pearson [1901]). This technique was also independently discovered by Harold Hotelling in 1933 (Hotelling [1933]). The idea behind PCA is mainly linear algebra. It first centers the data set at zero and then tries to find the largest eigenvalue of the covariance matrix. The corresponding eigenvector marks the direction of the most variance in the data, making it the biggest principle component. Then this component gets subtracted and the algorithm repeats to find the next largest eigenvalue. The algorithm can stop at any number of principle component. The corresponding eigenvalues and eigenvectors will then be used to construct a subspace with lower dimensions that represents the original data.

Intuitively speaking, PCA tries to fit a high-dimensional ellipsoid to the data set. Each axis of the ellipsoid represents one principle component. The larger the eigenvalue, the longer the corresponding axis of the ellipsoid. PCA is simple and elegant, so much so that it is still widely used today, but its shortcomings are also apparent. PCA assumes linearity as what it does is primarily finding a new coordinate system where the axes, a.k.a. the principle components, are linear combinations of the original variables. The mapping from the original high-dimensional space to the lower-dimensional subspace defined by principle components is an orthogonal linear transformation. This means that by analyzing the covariance matrix, PCA can mostly only capture linear dependencies. This approach will fail if the intrinsic structure of the data is non-linear. For example, PCA is likely to give two principle components if asked to analyze a circle, a typical 1-dimensional manifold in 2-dimensional Euclidean space. It is unable to see that the points on a circle are all part of a 1-dimensional space. Similar results with Swiss roll, PCA will very likely be unable to "unroll" the Swiss roll into its natural 2-dimensional representation, but rather cut through it.

PCA is one of the most important and widely used dimension reduction algorithms. Though its assumption on linearity limits its application, it also allows it to be computationally efficient using modern hardware and parallelism. It outperforms many modern algorithms in terms of speed and scalability. For this reason PCA is included in many modern software packages like scikit-learn and PyTorch.

Continuing onward, in the 1950s, Warren Torgerson invented Multidimensional Scaling (MDS) (Torgerson [1952], Torgerson [1958]), with the classical version often known as Torgerson Scaling. MDS is one of the early non-linear dimension reduction algorithms. Given a distance/dissimilarity matrix (where the distance does not have to be Euclidean distance) and a target dimension N , MDS aims to find a set of points in N -dimensional space where the pairwise distances are preserved as much as possible. Often one lets $N = 2$ so the result can be easily visualized. In classical MDS, the final goal is to minimize the loss function,

which is also called strain:

$$\text{Strain}(x_1, \dots, x_n) = \left(\frac{\sum_{i,j} (b_{ij} - x_i^T x_j)^2}{\sum_{i,j} b_{ij}^2} \right). \quad (2.4)$$

Here, x_i 's are vectors in N -dimensional space that we are trying to find, and b_{ij} are the entries from the Gram matrix that can be computed using the input data. Compared to PCA, MDS no longer assumes linear relationships between data points, and is more intuitive as the distance between points is often a good reflection of their similarity. One of the most famous applications of MDS is the color wheel. One might never hear about MDS, but they must have used the color wheel at some point of their lives.



Figure 1: The famous 2-dimensional color wheel, created by MDS. It effectively and intuitively captured the similarity and discrepancies between different colors on the spectrum of visible light.

Being intuitive and flexible, MDS gained a lot of popularity and it was developed into a family of algorithms. Shepard and Kruskal developed non-metric MDS independently in 1962 and 1964 (Shepard [1962], Kruskal [1964]). Their work allowed MDS to handle Ordinal data, meaning the modified algorithm only requires the rank order of dissimilarities. Such outstanding flexibility is one of the major reasons that it is still widely used today, especially when there is a lack of good distance metrics or quantitative data, such as market research. However, MDS is not without drawbacks. The axes of the output do not have inherent meanings from the original data set and sometimes it is not possible to find a solution with low enough stress values (the stain in the classical MDS). More importantly, MDS is not computationally efficient. Though modern algorithms like t-SNE and UMAP share some of the disadvantages with the MDS family, they often outperform MDS by a fair margin. Nevertheless, the limitations of MDS algorithms do not diminish their importance. MDS is still one of the most important algorithms in the field of dimension reduction.

Fast forwarding to the millennium, the rapid advancement of computer science and applied mathematics created increasingly stronger needs to study data of nonlinear nature. Scientists had been striving to create new tools and algorithms that can handle the emerging complex data sets. Though mentioned in earlier works, many regarded the work of Roweis and Saul in 2000 as the pillar of manifold learning (Roweis and Saul [2000]). Unlike PCA, which tries to project a collection of high dimensional vectors onto a linear subspace, manifold learning projects the high-dimensional data onto a lower-dimensional latent manifold. Since manifold learning was created specifically to tackle problems with nonlinear nature, it is also called Nonlinear Dimension Reduction (NLDR).

In the same paper, Roweis and Saul presented the Local Linear Embedding (LLE), one of the classic algorithms in NLDR. From this point onward, readers who are familiar with SNE and t-SNE may notice the resemblance between the algorithms and how they draw inspiration from each other. Given original data points $X_i \in \mathbb{R}^D$, the goal of LLE is to find a set of points $Y_i \in \mathbb{R}^d$ usually with $D \gg d$ such that Y_i 's form a good representation (embedding) of X_i 's. This is done in two steps. First, LLE looks at each point X_i and tries to find the closest approximation of X_i using the linear combination of other points. i.e., for each i , LLE computes

$$\left\| X_i - \sum_j W_{ij} X_j \right\|^2, \quad (2.5)$$

where W_{ij} are the weight we assigned to each X_j . W_{ij} is nonzero when X_j is within a certain distance of X_i , or when X_j is considered a "neighbor". This distance is a hyperparameter set by the user. Summing up through all i 's gives us the reconstruction error:

$$E(W) := \sum_{i=1}^N \left\| X_i - \sum_j W_{ij} X_j \right\|^2. \quad (2.6)$$

While LLE aims to minimize $E(W)$, it is also subject to the restriction

$$\sum_j W_{ij} = 1. \quad (2.7)$$

This ensures that W is not affected by rotation and translation. The first step is finished when LLE successfully finds the weight that minimizes $E(W)$. This is a typical constraint least square problem so we are able to find the global minimum. Now the weight W_{ij} 's are fixed and the second step begins. LLE now tries to find output Y_i 's that minimize the following cost function:

$$C(Y) := \sum_{i=1}^N \left\| Y_i - \sum_j W_{ij} Y_j \right\|^2. \quad (2.8)$$

The algorithm terminates when the minima of $C(Y)$ is found. Directly from the same paper (Roweis and Saul [2000]), we know that finding the minimum of $C(Y)$ is equivalent to solving a sparse eigenvalue problem and we are able to find the global minimum. This is one of the most appealing mathematical properties of LLE, that both of its core steps yield globally optimal solutions. However, nothing is perfect and LLE does come with its own shortcomings. Let us discuss that after we introduce another NLDR algorithm that was created around the same time, and is just as famous and important: Isomap.

2.3 The Road to t-SNE

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